PFP: Recursive re-entry with evidence scope and shared fuel

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Code baseline: 59dde95a7860063c25d75db37f40af556e9f28ee

Abstract

A returned answer, rejection rule, or operator can become the next proposal. This recursion is useful only when its task, evidence, transport, and resource bindings survive. We state a finite composition theorem, separate soundness from solution coverage, and exhibit a source-sound rejection rule that is unsound after a task change. Positive charged costs bound execution without establishing that the original problem has been solved.

Mechanism inventory

Mechanism	Evidence
Mathematical statement	Written proof under declared hypotheses
Finite example	Standard-library replay of submitted inputs and rejection controls
Model inference	Not executed
Lean verification	No new formalization claimed
Authorization	None supplied

1. The re-entry contract

Fix the original task P_0 and representations $P_1, \dots, P_n$. Each admitted edge supplies a backward solution map $b_i : \mathrm{Sol}(P_{i+1}) \to \mathrm{Sol}(P_i)$ and, separately, a forward coverage map f_i in the opposite direction. A certificate binds the source and target task data, checker versions, parent receipt, and operation charge. Assumptions cannot change silently. A changed assumption set produces a changed or conditional task result. An authorization requirement is supplied independently.

2. Finite composition theorem

Theorem. Under the declared edge obligations, a checked solution at P_n decodes to a solution at P_0 by $b_0 \circ \cdots \circ b_{n-1}$. Forward maps compose in the opposite order and retain the stated coverage. If original witness recovery is required, additionally assume $b_i \circ f_i = \mathrm{id}$; that law then composes. If every operation costs at least $\delta > 0$ and all prefixes charge one original budget B, at most $\lfloor B/\delta \rfloor$ operations execute.

Proof. Induct on history length. Apply the last backward map to the terminal solution, then the induction hypothesis. The forward argument is the same induction with arrows reversed. Cancel each adjacent $b_i f_i$ for the recovery law. For the cost bound, $n\delta \le \sum_i c_i \le B$. These are conditional composition laws: they do not construct missing adapters, prove checker soundness, or validate the legitimacy of a grant.

3. Soundness does not imply coverage

Let the original solution set be $\{0\}$ and the target solution set empty. The empty backward map is sound, but no forward coverage map exists. A target search can therefore fail despite an original solution. Conversely, an adapter preserving existence can still lose original witness identity unless the recovery law is supplied. These are different requirements.

4. Antibody transfer counterexample

An antibody is a rejection predicate r. Its validity for a task is $r(w) \Rightarrow w \notin \mathrm{Sol}(P)$. On the domain $\{0,1\}$, let $\mathrm{Sol}(P) = \{0\}$ and let r reject 1. It is source-sound. For a changed task P' with solution set $\{1\}$, the same predicate destroys the only valid answer. Transferring it requires a checked target validity condition or a transport theorem whose hypotheses apply to the target. Unchanged syntax and a familiar task name are insufficient.

Truth-sound rejection also need not preserve the reachability or cost of a controller strategy: a proposal channel may rely on intermediate candidates that are not terminal solutions. Appendix S supplies the additional game conditions for a declared finite execution. Task soundness, solution coverage, and protected operating budgets must be checked separately.

5. Re-entry example and replay

The companion checks a two-operation history with unit costs and total budget two. It rejects a stale parent, a changed task, and a child claiming to reset consumed fuel. It also checks the vacuous-soundness example and the antibody counterexample using complete finite sets. This is a reference contract example, not a claim that every engine adapter has these bindings. Model requests, installation, and external work require separately declared charges; omitting them weakens the resource statement.

6. Evidence boundary and contribution

Function composition and the cost inequality are standard mathematics. The PFP/WDP contribution is their placement in a retained re-entry contract with explicit task, parent, checker, and cost scope. The example is executable; the general statement is written. Constructing and independently checking the actual native re-entry adapter remains an open WDP target. Recursion alone does not guarantee resolution, intelligence, or authorization.

The two-level teaching packet includes shared system rules, this typed user instance, a different worked example and a required candidate response schema. It keeps provider system/user roles separate. It is a saved example, not a schema-admission or model-run receipt; the local replay checks the mathematical or protocol obligations stated above.

Research extension: explicit incumbent re-entry

The implemented research entrypoint makes a confirmed harness the parent used by the next outer proposal. Its state includes the original experiment and budget, current candidate bytes, parent identity, frozen checker and split identities, model configuration, and selection and confirmation receipts. A replay validates these bindings before reconstructing an incumbent chain. Ordinary learning hooks retain their recursion suppression; this explicit experimental path does not silently turn a successful completion notification into another provider call.

Candidate and incumbent are evaluated in paired assignments, followed by fresh confirmation. Exact completion fractions and recorded cost remain separate objectives. Correctness has priority, cost breaks resolved correctness ties, and an unresolved tie retains the incumbent. No task family may regress. Passing invariant controls is necessary but does not establish semantic correctness of all permitted future programs.

The shared-fuel argument above concerns positive charged costs under a fixed resource identity. The implemented ledger reserves maximum liability before transport and retains ambiguous charges across restart. A proposed child cannot reset the budget. This bounds authorized dispatch under the declared pricing and transport assumptions; it does not prove that a successor solves the original task or makes research more effective.

An empirical improvement receipt is not a backward solution map, a coverage map, or a proof of compositional soundness. Those objects still require the theorem's hypotheses. The three short paired ignition repetitions in Paper 029 are explicitly exploratory and cannot establish accelerating self-improvement. Current model performance is unmeasured; the new implementation extends the operational example without adding a formal theorem or altering the finite proof above.

The source and page references for this extension are collected in Paper 029 and its source-to-claim map. The author-review rendering does not replace a historical deposit or archive.

Sources and relation to the programme

This draft builds on the WDP manuscript, Paper 021's source-and-execution distinctions, and the corrected findings notices. Paper 022 supplies a research design whose live-model and offline-strategy claims require their own source-bound scrutiny. These drafts do not adopt its empirical tables as new measurements. The PFP collection guide provides the shared-system/typed-instance protocol. Genesis and MSS-N presentation stages do not license a new mathematical or physical carrier.